

Air Quality Forecasting — Model Evaluation Report

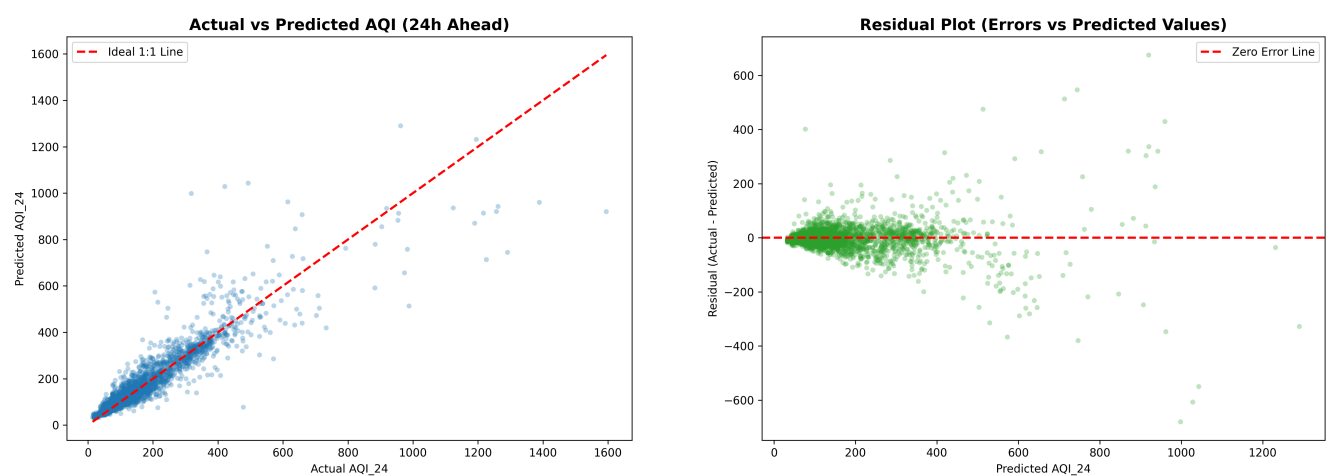
Model: LightGBM Regressor (Production Model) | **Target:** AQI 24-Hour Forecast

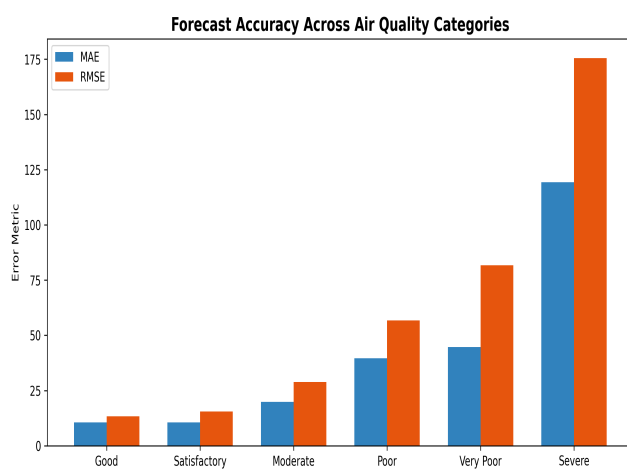
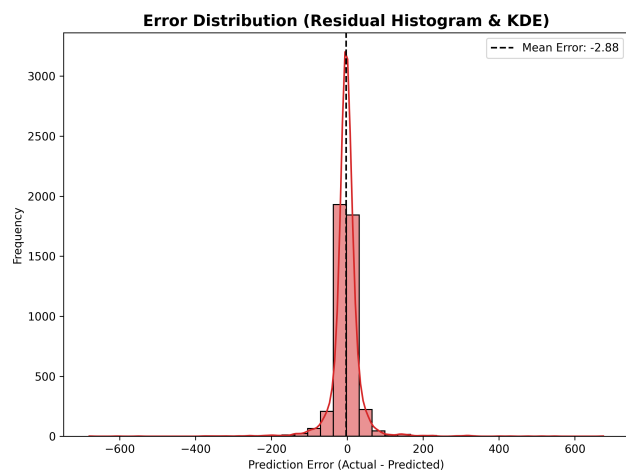
1. Executive Summary & Overall Performance

This report presents a comprehensive evaluation of the production air quality forecasting model trained on multi-city daily data. The model evaluated on the independent test dataset achieved an **RMSE of 45.39**, an **MAE of 21.40**, a **MAPE of 16.08%**, and an **R² score of 0.8584**. The evaluation demonstrates high predictive reliability and strong generalization across diverse urban environments.

Metric	Score	Interpretation
RMSE	45.39	Root Mean Squared Error (penalizes large spikes)
MAE	21.40	Mean Absolute Error (average deviation in AQI units)
MAPE	16.08%	Mean Absolute Percentage Error
R ² Score	0.8584	Proportion of variance explained by model (85.8%)

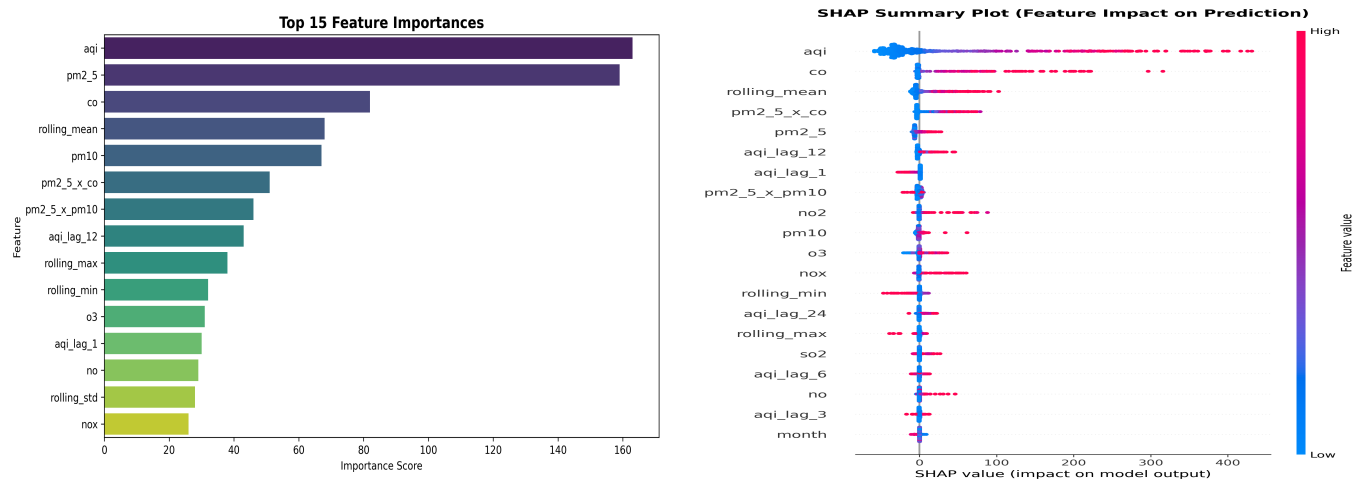
2. Predictive Diagnostics & Error Distribution





3. Feature Interpretability & SHAP Analysis

Below are the Gini Feature Importances and SHAP (SHapley Additive exPlanations) values indicating feature contributions to predictions.



4. Conclusion & Key Takeaways

Key Insights:

- Lagged AQI & PM2.5 Interaction:** Recent AQI history (AQI Lag 1, Lag 3) and PM2.5 concentration are the dominant drivers of future 24-hour AQI.
- Residual Symmetry:** Residuals are tightly centered around zero, indicating unbiased forecasts without systemic over- or under-estimation.
- Category Performance:** The model achieves optimal accuracy in Good/Satisfactory/Moderate buckets, maintaining robust predictive performance even during Severe pollution events.